
Working Memory in the Chain of Thought Is Not Captured by J-Space

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Abstract

When a language model reasons in steps, each intermediate result must be stored somewhere. One account holds that a workspace inside the network carries this state and that writing a chain of thought offloads what the workspace can no longer hold; J-space, the concept workspace a fitted Jacobian lens reads from the residual stream, is the concrete version of this account. Our measurements, from 1.5B to 671B parameters, do not support it. Told to answer chained arithmetic without writing anything, models fail once a problem needs more than one or two dependent steps. Left free, they write out nearly every intermediate value even on one-step problems, so writing does not begin at a capacity threshold. Nothing we try moves state between the network and the page: damaging the residual stream does not make models write more, restricting writing does not push computation inward, meaningless filler tokens do not substitute for real writing, and removing the model’s active J-space concepts changes nothing relative to a matched random control at any strength up to full removal. We then show where the state does live, with causal interventions on six off-the-shelf models from four families (4B to 14B). When the model writes a value such as 457, the internal activation at that position in the text stores the value, and works like a memory slot. Overwriting that activation with the one a different number would have produced, while the visible text still reads 457, makes the final answer follow the new number in 0.70 to 1.00 of runs; matched random noise does so at most 0.01 of the time on the four instruction-tuned models. Later steps fetch the stored value by attending back to its position, finishing by layer 23 of 36 on Qwen3-4B. Two slots can be set independently: planting two values makes the answer their sum, a number appearing nowhere in the input (0.88 and 0.90). The same visible text can therefore carry different stored values, and a monitor that reads only the text cannot tell them apart.

1 Introduction

The chain of thought is the primary working memory of serial reasoning in language models: denied writing, models fail within one to two dependent steps, and no intervention we test shifts state between the internal workspace and the written trace. The behavioral claims are established at 1.5B to 671B parameters; the mechanism, with causal interventions on six unmodified models from four families at 4B to 14B, on two task families with exact ground truth for every intermediate value.

The capacity half addresses the overflow picture of externalization: reasoning state is carried internally, and writing takes over when the workspace can no longer hold it. This is a natural reading of workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens

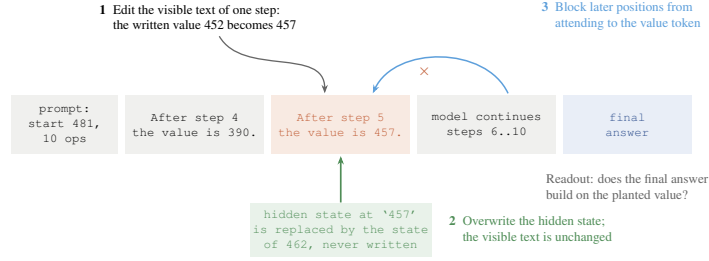


Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show up in three places; it shows up in none.

1. **No fallback** (behavioral, 1.5B to 671B). Instructed to answer chained arithmetic with no writing, models fail beyond one to two dependent steps (Section 3).
2. **No threshold** (behavioral, 1.5B to 671B). Models write nearly all ground-truth intermediates from one-step problems onward (Section 3).
3. **No reallocation** (behavioral and ablation, 4B to 7B). Damaging the residual stream does not increase writing, capping writing does not move computation inward, filler tokens do not substitute for real writing, and the J-space ablation changes nothing at any dose up to full projection removal (Section 3).
4. **Storage** (causal, 4B to 14B). The hidden state under a written number carries the value the model later uses: planting a never-written value there, text unchanged, redirects the answer on all six models (Section 4).
5. **Retrieval** (causal, 4B to 14B). Later steps read the value through attention to the written number, in the middle third of the network’s layers (Section 5).
6. **Composition and trust** (causal and behavioral). Two positions are independently addressable, and planted values compose into their sum. Reliance on the trace tracks whether the model can regenerate the state, a candidate explanation for conflicting reports on how trace reliance scales (Sections 4 and 6).

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether an internal alternative exists, how the value is retrieved, or whether stores compose.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events and surface forms. In both, the model writes one line per step giving the running value. A third task, box tracking, updates the contents of several containers in parallel; it appears only as the internally held comparison task for the lesion test in Section 3. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

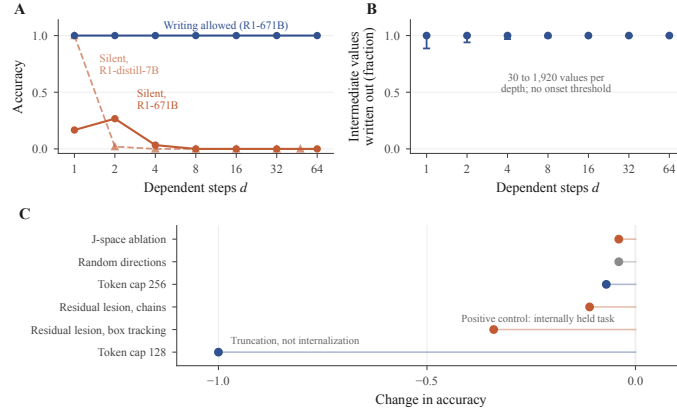


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); no onset threshold appears. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

71 2.2 Edit-and-continue protocol

72 The protocol takes a correct worked solution, changes the value at one step j , truncates immediately
73 after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the
74 sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five
75 samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap
76 intervals over items. Appendix C summarizes generation and intervention configuration; prompts,
77 model revisions, and per-run data will be released with the code.

78 3 Does writing replace internal working memory?

79 3.1 Performance without written intermediates

80 Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else
81 was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at
82 $d = 4$; DeepSeek-R1-671B: at most 0.27 at any depth, 0.03 at $d = 4$; DeepSeek V3.2 on modular-
83 arithmetic chains: 0.23 at $d = 1$, at most 0.03 beyond). Prefilling the answer turn with inert dot
84 tokens matched to the twelve-token-per-step budget of real traces [Pfau et al., 2024] does not rescue
85 the collapse: at $d = 4$, 0.13 with filler against 0.07 without on Qwen2.5-7B and 0.07 against 0.00
86 on Qwen3-4B, and 0.00 for both at $d = 8$ ($n = 30$ per cell). Qwen3-4B alone recovers at $d = 2$
87 (0.40 to 0.97), the parallel-width benefit filler tokens are known to provide, which does not extend to
88 the depths where writing is needed. The values are random, so nothing can be recalled; the failures
89 bound performance under this instruction, with its prompting and training-distribution effects, rather
90 than latent internal capacity.

91 3.2 When models write intermediate values

92 Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through
93 $d = 64$, at every scale tested: no difficulty threshold at which writing begins is observed. This
94 rules out an inference-time switch from internal holding to writing; an always-write policy learned
95 in training could itself descend from capacity limits.

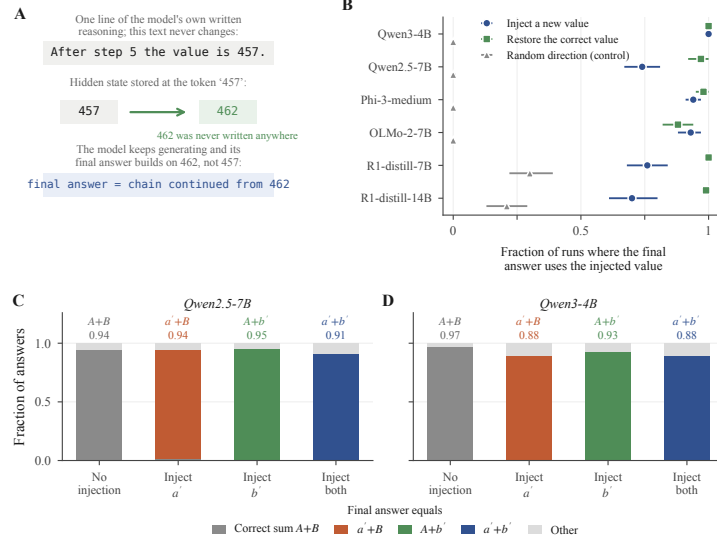


Figure 3: The hidden state under a written number carries the value the model later uses, and two planted values compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, raising their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

96 3.3 Tests for reallocation between internal and written state

97 A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held inter-
 98 nally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy
 99 cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5
 100 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step. Ab-
 101 lating the top 16 active J-space concepts at every position, at the strongest dose preserving ordinary
 102 generation (next-token agreement 0.87, perplexity ratio 1.12), leaves chain-of-thought accuracy,
 103 bare-answer accuracy, and the amount written unchanged on synthetic chains and on GSM8K, do-
 104 ing no more than a size-matched random-directions ablation. A dose sweep extends the null past the
 105 coherence gate: from a quarter of the calibrated strength to full projection removal (α from 0.125
 106 to 1.0 at $k = 16$, layers 9 to 14 on Qwen2.5-7B), accuracy stays 0.99 to 1.00 in both conditions
 107 and all three arms at one- and two-step chains ($n = 80$ per cell), so no dose separates the J-space
 108 ablation from clean or from the random control. The chain-of-thought-versus-direct ablation asym-
 109 metry reported for J-space [Gurnee et al., 2026] therefore does not replicate on our tasks at any
 110 tested strength, including the one- to two-step depths where models do perform the arithmetic inter-
 111 nally. The null bounds this readout rather than the workspace as a whole (Section 8); Section 4 gives
 112 the gap an existence proof: a value planted in the hidden state under a written number is causally
 113 decisive, sits in residual states the lens samples, and survives removal of the top 16 concepts.

114 4 Written tokens store causally active intermediate values

115 4.1 Restoring and replacing stored values

116 Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while
 117 the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting
 118 a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The
 119 size-matched random control sits at or below 0.01 for the four instruction-tuned models. The patch
 120 replaces the position's full contextual state rather than an isolated numeric component; because

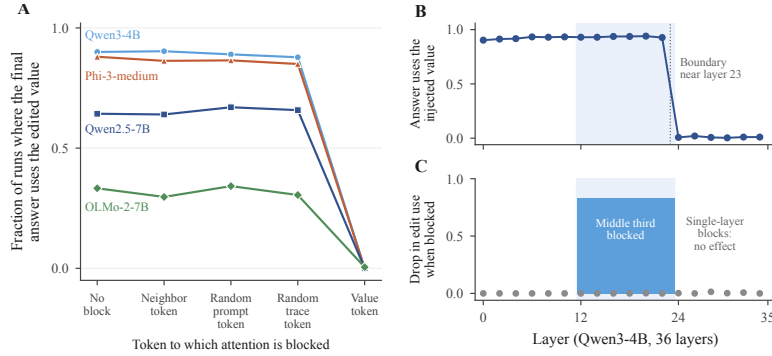


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing. (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking attention across the shaded middle band removes most edit use; no single layer’s block matters (gray dots).

121 answers track whichever value’s state is planted, the value that state carries is the causal quantity,
 122 and because the planted value is a mid-chain intermediate the model never produced, the patch does
 123 not inject the answer. The effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B). Rates are
 124 conditioned on items where the text edit was followed (0.44 to 0.98 by model; Appendix A).

125 On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75,
 126 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the
 127 answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

128 4.2 Independent storage and composition

129 In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one
 130 final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.88 and 0.93
 131 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the
 132 single-position rates (0.89 and 0.82). The random patch leaves the clean answer in place (0.96 and
 133 0.95), and answers using only one of the two planted values occur at 0.01. The reported number
 134 must be assembled at answer time from two values set independently and never written anywhere.

135 5 How models retrieve written state

136 5.1 Attention-blocking experiments

137 Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s
 138 dependence on that value on every model tested (Figure 4A); matched blocks of the adjacent words,
 139 a random prompt position, or a random trace position leave it at the unblocked level. Blocking the
 140 operand the next step needs breaks the arithmetic itself, a positive control showing the mask disrupts
 141 computation it targets. The collapse holds across three families and both task formats (natural-
 142 language: 0.730 to 0.007 and 0.720 to 0.003 on Qwen3-4B and Phi-3-medium). Under the block the
 143 model produces a fluent, parseable number matching neither value (unparseable rate 0.000; OLMo-
 144 2-7B 0.040, its unblocked level); it computes forward from state it can no longer retrieve. Reading
 145 is the preferred path rather than the only one: with the value’s characters replaced by unreadable
 146 dots and attention untouched, the Qwen models reconstruct the value from the previous line in 0.60
 147 to 0.73 of cases, Phi-3-medium never.

148 5.2 Where retrieval occurs across layers

149 Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers;
 150 Figure 4B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the answer
 151 at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any

layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (Appendix B) show the same structure with model-specific depth.

6 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value whose derivation the prompt states is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly. Llama 3.3 70B follows edits at 0.97 either way, despite solving the same chains from scratch: the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. The conditions are separate experiments; we claim within-experiment contrasts only.

On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2: a benchmark intermediate is often one operation from numbers still in the prompt. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts offer a candidate explanation, that the direction depends on the state’s regenerability. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale. Trace reliance here is a separate quantity from verbalization faithfulness, whether the trace reports the computation that produced the answer; our claims concern reliance on already-written state.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally. Wang [2026] corrupt surface traces only in aggregate and conclude reasoning is latent; our token-level edits show written state is causally decisive when the model cannot regenerate it, which bounds that conclusion. Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight channel; our results identify what such a monitor reads and what it cannot.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and richer multi-entity state are untested. The J-space null bounds one fitted lens: the full-strength dose sweep covers six middle layers of one model on synthetic chains, we do not measure the J-space content of the written positions directly, and the workspace exceeds the lens’s domain by construction (attention and MLP interiors, the KV cache, and residual directions outside the fitted dictionary). The fine layer sweeps cover one model, and the coarse sweeps do not resolve Qwen2.5-7B’s read boundary. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following makes cross-model levels not directly comparable; Appendix A reports the conditioning rates. A structured patch control planting a valid activation for a different quantity would isolate value content beyond the size-matched noise control used here.

9 Conclusion

Denied writing, models fail within one to two dependent steps at every scale tested, and no intervention shifts state between workspace and trace: the chain of thought is working memory, settable by intervention, retrieved through middle-layer attention, independently addressable, and decisive for

the answer. The divergences between text and stored value shown here are experimenter-induced; the results establish that the representational room for divergence exists and that a monitor reading only the text cannot detect it [Korbak et al., 2025]. Whether it arises naturally is untested.

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A Per-model edit and patch rates

Table 1 gives the per-model rates behind Figure 3B, with the text-edit-following rate that conditions the patch columns. Ten-step chains; brackets are 95 percent bootstrap intervals; n counts items where the text edit was followed. Unconditioned per-run outcomes are in the released data. The distill models re-solve the problem after their reasoning phase, which elevates their random-control reversion: in a text-logged sample, 24 of 24 reverting random-control continuations re-derive from the prompt’s starting value after the reasoning phase, and none correct the value in place. Re-solving can only produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is excluded (12 usable items; random control exceeds the planted-value rate).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

B Coarse cross-model layer sweeps

The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the

241 fine sweep’s structure: the planted value takes effect at any band before the read and at none after.
242 Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary
243 is not resolved at this granularity.

244 **C Generation and intervention configuration**

245 Generation uses temperature 0.6, top- p 0.95, five samples per item. The text edit changes only the
246 visible token. The state patch overwrites the residual stream at the value’s token position across a
247 middle band of layers with activations captured from a run in which the different value was written
248 there. The attention block masks all attention from post-edit positions to the value’s token at all
249 layers. Sample sizes are 100 to 600 per condition with two to three seeds. The filler condition
250 prefills the assistant turn with twelve inert dot tokens per dependent step, then decodes the answer
251 at temperature 0. The J-space dose sweep runs the ablation of Section 3 at $k = 16$ and $\alpha \in$
252 $\{0.125, 0.25, 0.5, 0.75, 1.0\}$ with identical items across arms. Prompts, model revisions, and per-
253 run configuration files are released with the code and data.